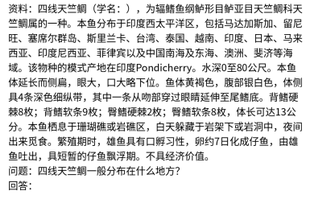
Native image-answer flow: trained execution, failed answering

a Actual input and sampling path

Input page x


资料：四线天竺鲷（学名：），为辐鳍鱼纲鲈形目鲈亚目天竺鲷科天竺鲷属的一种。本鱼分布于印度西太平洋区，包括马达加斯加、留尼旺、塞席尔群岛、斯里兰卡、台湾、泰国、越南、印度、日本、马来西亚、印度尼西亚、菲律宾以及中国南海及东海、澳洲、斐济等海域。该物种的模式产地在印度Pondicherry。水深0至80公尺。本鱼体延长而侧扁，眼大，口大略下位。鱼体黄褐色，腹部银白色，体侧具4条深色细纵带，其中一条从吻部穿过眼睛延伸至尾鳍底。背鳍硬棘8枚；背鳍软条9枚；臀鳍硬棘2枚；臀鳍软条8枚，体长可达13公分。本鱼栖息于珊瑚礁或岩礁区，白天躲藏于岩架下或岩洞中，夜间出来觅食。繁殖期时，雄鱼具有口孵习性，卵约7日化成仔鱼，由雄鱼吐出，具短暂的仔鱼飘浮期。不具经济价值。
问题：四线天竺鲷一般分布在什么地方？
回答：

Frozen image / language reader 3.443B parameters

Connector 0.437B parameters LoRA, rank 16

Image denoiser 2.276B parameters LoRA, rank 16

50 steps

Frozen VAE 0.084B parameters

Actual answer image



Full unrepaired page

1024 × 1024 RGB

Reference latent r(x): same frozen VAE encoder

b Training-only image supervision

Answer image y → frozen VAE encoder E(y)

$z\sigma = (1 - \sigma)\, E(y) + \sigma\, \varepsilon \qquad v^* = \varepsilon - E(y)$

$\text{Loss} = \text{MSE}(v\theta(z\sigma, t, x), v^*)$

One logical denoiser call per update; positive task branch.
Loss updates connector / writer adapters; reader and VAE stay frozen.

c Matched source / trained results

	Initial	Final
Seen questions	0/32	**0/32**
Held-out questions	0/32	**0/32**
Both held-out correct	0/16	**0/16**
Seen copying strings	0/32	**2/32**
Held-out copying strings	0/32	**3/32**

3.18 h training; 15.697 GiB peak
37.381M trained adapters; 6.240B retained runtime
QA reference OCR: 23/32 seen; 26/32 held out
Copy reference OCR: 27/32 seen; 29/32 held out

All 384 fixed initial/final outputs authenticated and inspected. Inherited language/image pretraining. Zero QA counts use source-accepted OCR; useful language learning remains unproved.